

Consider the incompressible Navier–Stokes equations on a domain Ω :

$$\partial_t u + (u \cdot \nabla)u + \nabla p - \nu \Delta u = f, \quad (1)$$

$$\nabla \cdot u = 0, \quad (2)$$

where $u(x, t) \in \mathbb{R}^d$ is velocity, $p(x, t) \in \mathbb{R}$ is pressure, $\nu > 0$ is kinematic viscosity, and f is a forcing term.

Define the (continuous) Navier–Stokes operator

$$\mathcal{N}(u, p) := (u \cdot \nabla)u + \nabla p - \nu \Delta u - f, \quad (3)$$

so (1) is equivalently

$$\partial_t u + \mathcal{N}(u, p) = 0, \quad \text{with } \nabla \cdot u = 0. \quad (4)$$

Let h denote a fine mesh size, and let $(u^h(t), p^h(t))$ be the fine-grid discrete solution (coming from your chosen spatial discretization: finite volume / finite difference / finite element):

$$\partial_t u^h + \mathcal{N}^h(u^h, p^h) = 0, \quad (5)$$

$$D^h u^h = 0. \quad (6)$$

Here:

- $\mathcal{N}^h$ is the discrete momentum operator (discrete advection + discrete pressure gradient + discrete diffusion $-f$),
- D^h is the discrete divergence operator.

Let H be a coarse mesh size ($H \gg h$). Introduce a restriction operator

$$R : \mathbb{R}^{n_h} \rightarrow \mathbb{R}^{n_H}, \quad (7)$$

mapping fine-grid velocity degrees of freedom to coarse-grid ones (e.g. cell-averaging, volume-weighted averaging, or filtering followed by sampling).

Define the coarse representation of the fine solution:

$$\bar{u}(t) := Ru^h(t), \quad \bar{p}(t) := R_p p^h(t) \quad (8)$$

(with R_p possibly different from R if pressure lives on different locations / spaces).

Differentiate $\bar{u} = Ru^h$ in time and use (5):

$$\partial_t \bar{u} = R \partial_t u^h = -R \mathcal{N}^h(u^h, p^h). \quad (9)$$

Thus the *true coarse-scale tendency induced by the fully resolved dynamics* is

$$-R \mathcal{N}^h(u^h, p^h). \quad (10)$$

Now discretize Navier–Stokes directly on the coarse mesh. Let $(u^H(t), p^H(t))$ satisfy:

$$\partial_t u^H + \mathcal{N}^H(u^H, p^H) = 0, \quad (11)$$

$$D^H u^H = 0. \quad (12)$$

To compare “truth restricted to coarse” with “coarse dynamics”, evaluate the coarse operator at the restricted state:

$$\mathcal{N}^H(\bar{u}, \bar{p}) = \mathcal{N}^H(Ru^h, R_p p^h). \quad (13)$$

Because Navier–Stokes is nonlinear (in particular the advection term), restriction and the nonlinear operator do *not* commute:

$$R \mathcal{N}^h(u^h, p^h) \neq \mathcal{N}^H(Ru^h, R_p p^h). \quad (14)$$

The **closure term** is defined as the discrepancy between:

- the restricted fine-grid tendency (truth projected to coarse),
- the coarse-grid operator applied to the restricted state.

Define

$$\mathcal{C}(\bar{u}, \bar{p}) := R \mathcal{N}^h(u^h, p^h) - \mathcal{N}^H(\bar{u}, \bar{p}) \quad (15)$$

(with $\bar{u} = Ru^h$, $\bar{p} = R_p p^h$).

Substitute (15) into (9):

$$\boxed{\partial_t \bar{u} + \mathcal{N}^H(\bar{u}, \bar{p}) = \mathcal{C}(\bar{u}, \bar{p})} \quad (16)$$

Equation (16) is the **exact coarse-scale equation**. It is “exact” in the sense that, if $\mathcal{C}$ were known and inserted, the evolution of $\bar{u}$ would match the restriction of the fine solution.

Write the discrete operators term-by-term on each grid:

$$\mathcal{N}^h(u^h, p^h) = \underbrace{\mathcal{A}^h(u^h)}_{\text{advection}} + \underbrace{\mathcal{G}^h p^h}_{\text{pressure gradient}} - \underbrace{\nu \mathcal{L}^h u^h}_{\text{diffusion}} - \underbrace{f^h}_{\text{forcing}}, \quad (17)$$

$$\mathcal{N}^H(\bar{u}, \bar{p}) = \underbrace{\mathcal{A}^H(\bar{u})}_{\text{advection}} + \underbrace{\mathcal{G}^H \bar{p}}_{\text{pressure gradient}} - \underbrace{\nu \mathcal{L}^H \bar{u}}_{\text{diffusion}} - \underbrace{f^H}_{\text{forcing}}. \quad (18)$$

Then the closure decomposes as

$$\mathcal{C} = \left(R\mathcal{A}^h(u^h) - \mathcal{A}^H(\bar{u}) \right) + \left(R\mathcal{G}^h p^h - \mathcal{G}^H \bar{p} \right) - \nu \left(R\mathcal{L}^h u^h - \mathcal{L}^H \bar{u} \right) - \left(Rf^h - f^H \right). \quad (19)$$

In most turbulent-flow contexts, the dominant contribution is typically the advection mismatch

$$\mathcal{C}_{\text{adv}} := R\mathcal{A}^h(u^h) - \mathcal{A}^H(\bar{u}), \quad (20)$$

because the nonlinearity strongly couples resolved and unresolved scales.

To connect with classical LES intuition, think of R as a filter (or averaging operator) that produces $\bar{u}$. For the continuous advection term:

$$\overline{(u \cdot \nabla)u} \neq (\bar{u} \cdot \nabla)\bar{u}. \quad (21)$$

The difference is commonly written as the divergence of a subgrid stress:

$$\overline{(u \cdot \nabla)u} = (\bar{u} \cdot \nabla)\bar{u} + \nabla \cdot \tau, \quad \tau := \overline{u \otimes u} - \bar{u} \otimes \bar{u}. \quad (22)$$

This τ term is precisely a closure: it depends on unresolved fluctuations and must be modeled.

Closure = (restricted fine-grid operator) – (coarse-grid operator on the restricted state).

(23)